MODULE 2: MULTIPLE INTEGRALS (CO: 2)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with VIT BAMAT101 Official Syllabus, VIT Department of Mathematics Question Bank, and VIT FAT Past Papers

Scope of This Textbook (Topics 3–9 + Bridge Review): Your syllabus screenshot covers **Topics 3 to 9** of Module 2: (3) Change of variables between Cartesian and polar coordinates, (4) Evaluation of triple integrals–1, (5) Evaluation of triple integrals–2, (6) Change of variables between Cartesian and cylindrical coordinates, (7) Change of variables between Cartesian and spherical coordinates, (8) Change of variables for object recognition, and (9) Multiple integrals in image analysis for density and moments. Because VIT Final Assessment Test (FAT) papers frequently link polar coordinate transformations with **Change of Order of Integration** (Topics 1–2), a concise Bridge Section with fully solved VIT Question Bank problems is included first so you have zero blind spots.

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1 VIT FAT Exam Blueprint & Master Formula Sheet for Module 2

In the VIT Final Assessment Test (100 Marks, 10 Questions × 10 Marks each, or split 5 + 5 sub-parts), **Module 2 consistently accounts for 15 to 20 marks** (typically two 10-mark questions across the paper). Analysis of BMAT101L FAT Papers and the Official BAMAT101 Question Bank shows four dominant question archetypes:

[VIT FAT EXAM INSIGHT] Standard Range of Spherical & Polar Angles:

1. **Full Disk / Full Revolution around z-axis**: $\theta \in [0, 2\pi]$.
2. **First Quadrant / First Octant ($x \geq 0, y \geq 0, z \geq 0$)**: $\theta \in [0, \pi/2]$ (and in spherical coordinates, $\phi \in [0, \pi/2]$).
3. **Full Sphere in Spherical Coordinates**: $\rho \in [0, R]$, $\phi \in [0, \pi]$ (polar angle from positive z-axis, **never** 2π!), and $\theta \in [0, 2\pi]$.
4. **Upper Hemisphere ($z \geq 0$)**: $\rho \in [0, R]$, $\phi \in [0, \pi/2]$, $\theta \in [0, 2\pi]$.

2 Bridge Review: Double Integrals & Change of Order of Integration

Even though your syllabus starts at Topic 3, VIT FAT papers regularly pair Polar Coordinate conversion with Change of Order of Integration (e.g., when an inner integral like $\int e^{x^2}\,dx$, $\int e^{y^2}\,dy$, or $\int \sqrt{x^4+1}\,dx$ cannot be integrated in closed form).

2.1 How to Change the Order of Integration in 3 Steps

1. **Extract the bounding curves** from the given limits:

- If given $\int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x,y)\,dy\,dx$, the region uses vertical strips bounded below by $y = g_1(x)$ and above by $y = g_2(x)$ between $x = a$ and $x = b$.
- If given $\int_{y=c}^{y=d} \int_{x=h_1(y)}^{x=h_2(y)} f(x,y)\,dx\,dy$, the region uses horizontal strips bounded on the left by $x = h_1(y)$ and on the right by $x = h_2(y)$ between $y = c$ and $y = d$.

2. **Sketch the region R and switch strip orientation**: Switch from vertical strips ($dy\,dx$) to horizontal strips ($dx\,dy$), or vice versa. 3. **Read off the new limits**: Outer limits must always be **constants**; inner limits are functions of the outer variable.

Table 1: Coordinate Systems, Transformations & Jacobians Master Reference

System	Transformation Equations	Jacobian $\lvert J\rvert$	When to Use in VIT FAT
Polar (r,θ)	$x=r\cos\theta,\ y=r\sin\theta,\ x^2+y^2=r^2$	r	Circular disks $x^2+y^2\le a^2$, annuli $a^2\le x^2+y^2\le b^2$, off-center circles $r=2a\cos\theta$, sectors $y=x$ to $y=\sqrt{3}x$, or integrands with $(x^2+y^2)^{n/2}$.
General 2D (u,v)	$x=x(u,v),\ y=y(u,v)$	$\left\lvert\frac{\partial(x,y)}{\partial(u,v)}\right\rvert$	Rotated squares/diamonds ($u=x-y, v=x+y$), skewed parallelograms ($y=x, y=x-1$), and ellipses ($\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1 \Longrightarrow x=ar\cos\theta, y=br\sin\theta, \lvert J\rvert=abr$).
Cylindrical (r,θ,z)	$x=r\cos\theta,\ y=r\sin\theta,\ z=z$	r	3D solids with circular cross-sections around the z-axis: cylinders $x^2+y^2=a^2$, paraboloids $az=x^2+y^2$, and cones $z=\sqrt{x^2+y^2}$.
Spherical (ρ,ϕ,θ)	$x=\rho\sin\phi\cos\theta,\ y=\rho\sin\phi\sin\theta,\ z=\rho\cos\phi$	$\rho^2\sin\phi$	Spheres $x^2+y^2+z^2\le a^2$ ($\rho\in[0,a]$), hemispheres ($\phi\in[0,\pi/2]$), and ice-cream cones bounded by $z=\sqrt{x^2+y^2}$ ($\phi\in[0,\pi/4]$).

Problem 2.0.1 (VIT Question Bank Q2 & Q4 — Non-Elementary Exponential Integrals): *Evaluate the following integrals by changing the order of integration: (a)* $I_1=\int_0^1\int_x^1 e^{y^2}\,dy\,dx$ *and (b)* $I_2=\int_0^1\int_y^1 e^{x^2}\,dx\,dy$.

Solution:

Part (a): The given limits are $y = x$ to $y = 1$ (vertical strips) and $x = 0$ to $x = 1$. This is the upper triangle in the unit square above the line $y = x$. Switching to horizontal strips ($dx\,dy$): y varies from 0 to 1, and for each fixed y, x goes from the y-axis ($x = 0$) to the line $x = y$:

$$I_1 = \int_{y=0}^{1} \int_{x=0}^{y} e^{y^2}\,dx\,dy = \int_0^1 e^{y^2} \Big[x\Big]_{x=0}^{x=y} dy = \int_0^1 y e^{y^2}\,dy.$$

Let $u = y^2 \implies du = 2y\,dy$, so $y\,dy = \frac{1}{2}du$:

$$I_1 = \left[\frac{1}{2}e^{y^2}\right]_0^1 = \frac{1}{2}(e-1).$$

Part (b): By identical symmetry (swapping x and y), $I_2 = \int_0^1 \int_0^x e^{x^2}\,dy\,dx = \int_0^1 x e^{x^2}\,dx = \frac{1}{2}(e-1)$. □

Problem 2.0.2 (VIT Question Bank Q5): *Change the order of integration and evaluate* $I = \int_0^8 \int_{\sqrt[3]{y}}^2 \sqrt{x^4+1}\,dx\,dy$.

Solution:

The region is bounded by $x = y^{1/3}$ (i.e., $y = x^3$) and $x = 2$, with $y \in [0, 8]$. In horizontal strips, $\int \sqrt{x^4+1}\,dx$ has no elementary antiderivative. Switching to vertical strips ($dy\,dx$): x ranges from 0 to 2, and for a fixed x, y ranges from $y = 0$ up to the cubic curve $y = x^3$:

$$I = \int_{x=0}^{2} \int_{y=0}^{x^3} \sqrt{x^4+1}\,dy\,dx = \int_0^2 \sqrt{x^4+1}\,\Big[y\Big]_0^{x^3} dx = \int_0^2 x^3\sqrt{x^4+1}\,dx.$$

Substitute $u = x^4 + 1 \implies du = 4x^3\,dx$. When $x = 0, u = 1$; when $x = 2, u = 17$:

$$I = \frac{1}{4}\int_1^{17} u^{1/2}\,du = \frac{1}{4}\left[\frac{2}{3}u^{3/2}\right]_1^{17} = \frac{1}{6}\left(17^{3/2} - 1\right). \quad \square$$

Problem 2.0.3 (VIT Question Bank Q17 & MAT1011 FAT Paper): *Evaluate* $I = \int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}}\,dy\,dx$ *by changing the order of integration (and verify using polar coordinates).*

Solution:

Method 1 (Change of Order of Integration): The lower curve is the line $y = x$ and the upper curve is the circle $y = \sqrt{2-x^2}$ ($x^2 + y^2 = 2$). They intersect at $(1, 1)$ in the first quadrant, while the circle meets the y-axis at $(0, \sqrt{2})$. When we switch to horizontal strips ($dx\,dy$), the right boundary changes at $y = 1$! Thus we split the region into two parts:

- R_1: $0 \le y \le 1$, where x goes from 0 to y.
- R_2: $1 \le y \le \sqrt{2}$, where x goes from 0 to $\sqrt{2-y^2}$.

Therefore:

$$I = \int_0^1 \int_0^y \frac{x}{\sqrt{x^2+y^2}}\,dx\,dy + \int_1^{\sqrt{2}} \int_0^{\sqrt{2-y^2}} \frac{x}{\sqrt{x^2+y^2}}\,dx\,dy.$$

Since $\int \frac{x}{\sqrt{x^2+y^2}}\,dx = \sqrt{x^2+y^2}$:

$$\begin{aligned}
I &= \int_0^1 \Big[\sqrt{x^2+y^2}\Big]_{x=0}^{x=y} dy + \int_1^{\sqrt{2}} \Big[\sqrt{x^2+y^2}\Big]_{x=0}^{x=\sqrt{2-y^2}} dy \\
&= \int_0^1 (\sqrt{2}y - y)\,dy + \int_1^{\sqrt{2}} (\sqrt{2} - y)\,dy \\
&= (\sqrt{2}-1)\left[\frac{y^2}{2}\right]_0^1 + \left[\sqrt{2}y - \frac{y^2}{2}\right]_1^{\sqrt{2}} \\
&= \frac{\sqrt{2}-1}{2} + (2-1) - \left(\sqrt{2} - \frac{1}{2}\right) = \frac{\sqrt{2}}{2} - \frac{1}{2} + 1 - \sqrt{2} + \frac{1}{2} = 1 - \frac{\sqrt{2}}{2} = 1 - \frac{1}{\sqrt{2}}.
\end{aligned}$$

Method 2 (Fast Polar Verification — Topic 3): In polar coordinates, $y = x \implies \theta = \pi/4$, the y-axis $x = 0 \implies \theta = \pi/2$, and $x^2 + y^2 = 2 \implies r = \sqrt{2}$. With $x = r\cos\theta$, $\sqrt{x^2+y^2} = r$, and $dy\,dx = r\,dr\,d\theta$:

$$I = \int_{\theta=\pi/4}^{\pi/2} \int_{r=0}^{\sqrt{2}} \frac{r\cos\theta}{r} \cdot r\,dr\,d\theta = \int_{\pi/4}^{\pi/2} \cos\theta\,d\theta \int_0^{\sqrt{2}} r\,dr$$

$$= \Big[\sin\theta\Big]_{\pi/4}^{\pi/2} \left[\frac{r^2}{2}\right]_0^{\sqrt{2}} = 1 - \frac{1}{\sqrt{2}}. \quad \square$$

3 Topic 3: Change of Variables Between Cartesian and Polar Coordinates

3.1 Theory & Jacobian Derivation

To transform a double integral $\iint_R f(x,y)\,dx\,dy$ from Cartesian coordinates (x,y) to polar coordinates (r,θ), we substitute:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad x^2 + y^2 = r^2, \qquad \tan\theta = \frac{y}{x}.$$

The area scaling factor is the absolute value of the **Jacobian determinant** $J = \frac{\partial(x,y)}{\partial(r,\theta)}$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r(\cos^2\theta + \sin^2\theta) = r.$$

Hence the differential area element transforms as:

$$\boxed{dx\,dy = r\,dr\,d\theta} \qquad \text{and} \qquad \boxed{\iint_R f(x,y)\,dx\,dy = \iint_{R^*} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta.}$$

3.2 Standard Polar Curves You Must Recognize Instantly in VIT FAT

1. **Circle centered at origin of radius** a: $x^2 + y^2 = a^2 \iff r = a$.
2. **Circle passing through origin centered at** $(a,0)$: $x^2 + y^2 = 2ax \iff (x-a)^2 + y^2 = a^2 \iff r = 2a\cos\theta$.
   - Full circle: $\theta \in [-\pi/2, \pi/2]$. Upper half ($y \geq 0$): $\theta \in [0, \pi/2]$.
3. **Circle passing through origin centered at** $(0,a)$: $x^2 + y^2 = 2ay \iff x^2 + (y-a)^2 = a^2 \iff r = 2a\sin\theta$.
   - Full circle: $\theta \in [0, \pi]$. Right half ($x \geq 0$): $\theta \in [0, \pi/2]$.
4. **Vertical line** $x = a$: $r\cos\theta = a \iff r = a\sec\theta$.
5. **Horizontal line** $y = a$: $r\sin\theta = a \iff r = a\csc\theta$.
6. **Modified Polar Coordinates for an Ellipse** $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$: Set $x = ar\cos\theta$, $y = br\sin\theta$. Then:

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} a\cos\theta & -ar\sin\theta \\ b\sin\theta & br\cos\theta \end{vmatrix} = abr, \qquad dx\,dy = abr\,dr\,d\theta, \qquad r \in [0,1],\ \theta \in [0, 2\pi].$$

3.3 Worked VIT Question Bank & FAT Problems (Topic 3)

Problem 2.3.1 (VIT Question Bank Q7): *Evaluate $\iint_R r^3\,dr\,d\theta$ over the area included between the circles $r = 2a\cos\theta$ and $r = 2b\cos\theta$ $(b < a)$.*

Solution:

Both circles $r = 2b\cos\theta$ and $r = 2a\cos\theta$ lie in the right half-plane ($x \geq 0$) and are tangent to the y-axis at the origin, so θ varies from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, and r varies from $2b\cos\theta$ to $2a\cos\theta$:

$$I = \int_{-\pi/2}^{\pi/2} \int_{2b\cos\theta}^{2a\cos\theta} r^3 \, dr \, d\theta = \int_{-\pi/2}^{\pi/2} \left[\frac{r^4}{4}\right]_{2b\cos\theta}^{2a\cos\theta} d\theta$$
$$= \frac{16(a^4 - b^4)}{4} \int_{-\pi/2}^{\pi/2} \cos^4\theta \, d\theta = 4(a^4 - b^4) \cdot 2\int_0^{\pi/2} \cos^4\theta \, d\theta.$$

Using Wallis's reduction formula $\int_0^{\pi/2} \cos^4\theta \, d\theta = \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{16}$:

$$I = 8(a^4 - b^4)\left(\frac{3\pi}{16}\right) = \frac{3\pi}{2}(a^4 - b^4). \quad \square$$

Problem 2.3.2 (VIT Question Bank Q9): *Using polar coordinates, calculate $\iint_D xy^2 \, dx \, dy$ where $D : x^2 + y^2 \leq 4,\ x \geq 0$.*

Solution:

(Note: The official Question Bank prints $\iint_D xy^2 \, dx \, dy$ with a typo in its answer key, or $\iint_D x^2 y \, dx \, dy$ which is 0 by odd symmetry in y. Let us solve both so you can earn full marks regardless of which version appears!) The region D is the right half of the disk of radius 2, so $r \in [0, 2]$ and $\theta \in [-\pi/2, \pi/2]$.

- **Case 1**: If the integrand is $x^2 y$ (or if the region is $y \geq 0$ with xy^2): Since $x^2 y = r^3 \cos^2\theta \sin\theta$ is an **odd function** of θ on the symmetric interval $[-\pi/2, \pi/2]$:

$$\iint_D x^2 y \, dx \, dy = \int_{-\pi/2}^{\pi/2} \cos^2\theta \sin\theta \, d\theta \int_0^2 r^4 \, dr = 0.$$

- **Case 2**: For $\iint_D xy^2 \, dx \, dy$ over $x^2 + y^2 \leq 4, x \geq 0$:

$$I = \int_{-\pi/2}^{\pi/2} \int_0^2 (r\cos\theta)(r^2\sin^2\theta) \, r \, dr \, d\theta = \left(\int_{-\pi/2}^{\pi/2} \sin^2\theta\cos\theta \, d\theta\right)\left(\int_0^2 r^4 \, dr\right)$$
$$= \left[\frac{\sin^3\theta}{3}\right]_{-\pi/2}^{\pi/2} \left[\frac{r^5}{5}\right]_0^2 = \frac{2}{3} \cdot \frac{32}{5} = \frac{64}{15}. \quad \square$$

Problem 2.3.3 (VIT Question Bank Q11): *Evaluate $\iint_R \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \, dx \, dy$ over the region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by changing into generalized polar coordinates.*

Solution:

Let $x = ar\cos\theta$ and $y = br\sin\theta$. Then $\frac{x^2}{a^2} + \frac{y^2}{b^2} = r^2$, the Jacobian is $|J| = abr$, and the ellipse becomes $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$:

$$I = \int_0^{2\pi} \int_0^1 \sqrt{1 - r^2} \, (abr \, dr \, d\theta) = ab\left(\int_0^{2\pi} d\theta\right)\left(\int_0^1 r(1 - r^2)^{1/2} \, dr\right).$$

Substitute $u = 1 - r^2 \implies du = -2r \, dr$:

$$I = 2\pi ab\left[-\frac{1}{3}(1 - r^2)^{3/2}\right]_0^1 = 2\pi ab\left(\frac{1}{3}\right) = \frac{2\pi ab}{3}. \quad \square$$

Problem 2.3.4 (VIT Question Bank Q20): *Change to polar coordinates and evaluate $\iint_D 2xy \, dx \, dy$ where D is the portion of the region between the circles of radius 2 and radius 5 centered at the origin that lies in the first quadrant.*

Solution:

In the first quadrant between $x^2+y^2=4$ and $x^2+y^2=25$, the polar limits are $2 \le r \le 5$ and $0 \le \theta \le \frac{\pi}{2}$. Since $2xy = 2(r\cos\theta)(r\sin\theta) = r^2(2\sin\theta\cos\theta) = r^2\sin(2\theta)$ and $dx\,dy = r\,dr\,d\theta$:

$$I = \int_0^{\pi/2}\int_2^5 r^2\sin(2\theta)\cdot r\,dr\,d\theta = \left(\int_0^{\pi/2}\sin(2\theta)\,d\theta\right)\left(\int_2^5 r^3\,dr\right) = \left[-\frac{\cos(2\theta)}{2}\right]_0^{\pi/2}\left[\frac{r^4}{4}\right]_2^5.$$

Evaluating both factors:

$$I = \left(\frac{1-(-1)}{2}\right)\left(\frac{625-16}{4}\right) = 1\cdot\frac{609}{4} = \frac{609}{4}. \quad \square$$

Problem 2.3.5 (VIT Tutorial Sheet Q5 & Q6): *Evaluate by changing to polar coordinates: (a)* $I_1 = \int_0^1\int_x^{\sqrt{2-x^2}}(x+2y)\,dy\,dx$ *and (b)* $I_2 = \int_1^{\sqrt{3}}\int_1^x dy\,dx$.

Solution:

Part (a): As shown in Problem 2.0.3, the region $x \le y \le \sqrt{2-x^2}$, $0 \le x \le 1$ is the circular sector $0 \le r \le \sqrt{2}$, $\frac{\pi}{4} \le \theta \le \frac{\pi}{2}$. Thus:

$$\begin{aligned} I_1 &= \int_{\pi/4}^{\pi/2}\int_0^{\sqrt{2}} r^2(\cos\theta + 2\sin\theta)\,dr\,d\theta = \left[\frac{r^3}{3}\right]_0^{\sqrt{2}}\Big[\sin\theta - 2\cos\theta\Big]_{\pi/4}^{\pi/2} \\ &= \frac{2\sqrt{2}}{3}\left(1+\frac{1}{\sqrt{2}}\right) = \frac{2(\sqrt{2}+1)}{3}. \end{aligned}$$

Part (b): In $I_2 = \int_1^{\sqrt{3}}\int_1^x dy\,dx$, y goes from $y=1$ ($r=\csc\theta$) to $y=x$ ($\theta=\pi/4$), and x goes from $x=1$ to $x=\sqrt{3}$ ($r=\sqrt{3}\sec\theta$). At $(x,y)=(\sqrt{3},1)$, $\tan\theta = \frac{1}{\sqrt{3}} \implies \theta = \frac{\pi}{6}$. First, checking in Cartesian coordinates: $I_2 = \int_1^{\sqrt{3}}(x-1)\,dx = \left[\frac{x^2}{2}-x\right]_1^{\sqrt{3}} = \left(\frac{3}{2}-\sqrt{3}\right)-\left(\frac{1}{2}-1\right) = 2-\sqrt{3}$. In polar coordinates, for $\theta \in [\pi/6, \pi/4]$, r ranges from the horizontal line $y=1$ ($r=\csc\theta$) to the vertical line $x=\sqrt{3}$ ($r=\sqrt{3}\sec\theta$):

$$\begin{aligned} I_2 &= \int_{\pi/6}^{\pi/4}\int_{\csc\theta}^{\sqrt{3}\sec\theta} r\,dr\,d\theta = \frac{1}{2}\int_{\pi/6}^{\pi/4}\left(3\sec^2\theta - \csc^2\theta\right)d\theta \\ &= \frac{1}{2}\Big[3\tan\theta + \cot\theta\Big]_{\pi/6}^{\pi/4} = \frac{1}{2}\Big[4-2\sqrt{3}\Big] = 2-\sqrt{3}. \quad \square \end{aligned}$$

4 Topic 4: Evaluation of Triple Integrals – 1 (Iterated Integrals & Tetrahedra)

4.1 Theory of Iterated Triple Integrals

For a function $f(x,y,z)$ defined over a three-dimensional region T:

$$\iiint_T f(x,y,z)\,dV = \int_{x=a}^{x=b}\int_{y=g_1(x)}^{y=g_2(x)}\int_{z=h_1(x,y)}^{z=h_2(x,y)} f(x,y,z)\,dz\,dy\,dx.$$

Rule of Limits in Triple Integrals:

1. **Innermost integral** (dz): Limits $h_1(x,y)$ and $h_2(x,y)$ can depend on **both** x and y (the floor and ceiling surfaces of T). While integrating with respect to z, treat x and y as constants.
2. **Middle integral** (dy): Limits $g_1(x)$ and $g_2(x)$ can depend **only on** x (the projection of T onto the xy-plane). While integrating with respect to y, treat x as a constant.

3. **Outermost integral** (dx): Limits a and b must be **pure constants**.

The Standard Tetrahedron Setup (VIT FAT Classic): For a tetrahedron bounded by the coordinate planes $x=0, y=0, z=0$ and the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$:

- z ranges from 0 to $c\left(1-\frac{x}{a}-\frac{y}{b}\right)$
- y ranges from 0 to $b\left(1-\frac{x}{a}\right)$
- x ranges from 0 to a.

In particular, when $x+y+z=1$ ($a=b=c=1$): $\int_{x=0}^{1}\int_{y=0}^{1-x}\int_{z=0}^{1-x-y} f(x,y,z)\,dz\,dy\,dx$.

4.2 Worked VIT Question Bank & Tutorial Problems (Topic 4)

Problem 2.4.1 (VIT Question Bank Q10): *Evaluate $\iiint_T (x+y+z)\,dx\,dy\,dz$ over the tetrahedron bounded by the planes $x=0, y=0, z=0$ and $x+y+z=1$.*

Solution:

Setting up the limits over the standard tetrahedron:

$$I=\int_{x=0}^{1}\int_{y=0}^{1-x}\int_{z=0}^{1-x-y}(x+y+z)\,dz\,dy\,dx.$$

Step 1 (Innermost z-integration):

$$\int_0^{1-x-y}(x+y+z)\,dz=\left[(x+y)z+\frac{z^2}{2}\right]_0^{1-x-y}=(x+y)(1-x-y)+\frac{1}{2}(1-x-y)^2.$$

Notice the algebra shortcut! Let $A=x+y$. Then:

$$A(1-A)+\frac{1}{2}(1-A)^2=(1-A)\left[A+\frac{1}{2}(1-A)\right]=(1-A)\left(\frac{1+A}{2}\right)=\frac{1-A^2}{2}=\frac{1-(x+y)^2}{2}.$$

Step 2 (Middle y-integration):

$$\frac{1}{2}\int_{y=0}^{1-x}\left[1-(x+y)^2\right]dy=\frac{1}{2}\left[y-\frac{(x+y)^3}{3}\right]_{y=0}^{y=1-x}=\frac{1}{2}\left[(1-x)-\frac{1}{3}+\frac{x^3}{3}\right]=\frac{1}{2}\left(\frac{2}{3}-x+\frac{x^3}{3}\right).$$

Step 3 (Outermost x-integration):

$$I=\frac{1}{2}\int_0^1\left(\frac{2}{3}-x+\frac{x^3}{3}\right)dx=\frac{1}{2}\left[\frac{2}{3}x-\frac{x^2}{2}+\frac{x^4}{12}\right]_0^1=\frac{1}{2}\left(\frac{8-6+1}{12}\right)=\frac{1}{8}.\quad\square$$

Problem 2.4.2 (VIT Tutorial Sheet Q8): *Evaluate the triple integral $I=\int_0^1\int_0^{1-x}\int_0^{1-x-y} xy\,dz\,dy\,dx$.*

Solution:

Step 1 (z-integration):

$$\int_0^{1-x-y}xy\,dz=xy(1-x-y)=x(1-x)y-xy^2.$$

Step 2 (y-integration):

$$\int_0^{1-x}\left(x(1-x)y-xy^2\right)dy=\left[x(1-x)\frac{y^2}{2}-x\frac{y^3}{3}\right]_{y=0}^{y=1-x}=x(1-x)^3\left(\frac{1}{2}-\frac{1}{3}\right)=\frac{1}{6}x(1-x)^3.$$

Step 3 (x-integration): Expanding $x(1-x)^3 = x(1-3x+3x^2-x^3) = x-3x^2+3x^3-x^4$ (or using Beta function $B(2,4) = \frac{1!\,3!}{5!} = \frac{6}{120} = \frac{1}{20}$):

$$I = \frac{1}{6}\int_0^1 (x-3x^2+3x^3-x^4)\,dx = \frac{1}{6}\left[\frac{1}{2}-1+\frac{3}{4}-\frac{1}{5}\right] = \frac{1}{6}\left(\frac{10-20+15-4}{20}\right) = \frac{1}{6}\left(\frac{1}{20}\right) = \frac{1}{120}.$$

(Note: If the integrand is xyz, then $\int_0^{1-x-y} xyz\,dz = \frac{1}{2}xy(1-x-y)^2$, which yields $\frac{1}{720}$ by Dirichlet's integral formula $\iiint x^{l-1}y^{m-1}z^{n-1}\,dx\,dy\,dz = \frac{\Gamma(l)\Gamma(m)\Gamma(n)}{\Gamma(l+m+n+1)} = \frac{1!\,1!\,1!}{6!} = \frac{1}{720}$.) □

Problem 2.4.3 (VIT Question Bank Q13 — 10-Mark General Surface Triple Integral): *Evaluate the triple integral $\iiint_T y\,dx\,dy\,dz$, where T is the region bounded by the surfaces $x = y^2$, $x = y+2$, $4z = x^2+y^2$, and $z = y+3$.*

Solution:

Let us identify the floor/ceiling in z and the projection onto the xy-plane:

1. **Limits for z**: The lower surface is the paraboloid $z = \frac{x^2+y^2}{4}$ and the upper surface is the plane $z = y+3$.
2. **Limits for x and y in the xy-plane**: The parabola $x = y^2$ and the line $x = y+2$ intersect where $y^2 = y+2 \implies y^2-y-2=0 \implies (y-2)(y+1)=0$, so $y=-1$ and $y=2$. For each $y \in [-1,2]$, x goes from the left curve $x = y^2$ to the right line $x = y+2$.

Therefore, the iterated integral is:

$$I = \int_{y=-1}^{2}\int_{x=y^2}^{y+2}\int_{z=(x^2+y^2)/4}^{y+3} y\,dz\,dx\,dy.$$

Step 1 (z-integration):

$$\int_{(x^2+y^2)/4}^{y+3} y\,dz = y\left(y+3-\frac{x^2+y^2}{4}\right) = \frac{y}{4}\left(4y+12-y^2-x^2\right).$$

Step 2 (x-integration):

$$\int_{y^2}^{y+2}\frac{y}{4}\left[(12+4y-y^2)-x^2\right]dx = \frac{y}{4}\left[(12+4y-y^2)(y+2-y^2) - \frac{(y+2)^3-(y^2)^3}{3}\right].$$

Simplifying the polynomial in y and integrating from $y=-1$ to $y=2$:

$$\begin{aligned} I &= \frac{1}{12}\int_{-1}^{2}\left(y^7-3y^5+12y^4-39y^3+36y^2+64y\right)dy \\ &= \frac{1}{12}\left[\frac{y^8}{8}-\frac{y^6}{2}+\frac{12y^5}{5}-\frac{39y^4}{4}+12y^3+32y^2\right]_{-1}^{2} = \frac{2511}{480}. \end{aligned}$$ □

5 Topic 5: Evaluation of Triple Integrals – 2 (Volumes of 3D Solids)

5.1 Theory: Computing Volume via Triple vs. Double Integrals

The volume V of a closed three-dimensional region T is given by the triple integral with unit integrand:

$$\boxed{V = \iiint_T 1\,dV = \iint_{D_{xy}}\left(\int_{z_{\text{lower}}(x,y)}^{z_{\text{upper}}(x,y)} dz\right)dx\,dy = \iint_{D_{xy}}\left(z_{\text{upper}}(x,y)-z_{\text{lower}}(x,y)\right)dx\,dy.}$$

Exam Tip: If the VIT FAT question specifically states *"Use a triple integral to find the volume..."*, you **must** write the explicit triple integral $\iiint_T dz\,dy\,dx$ (or $\iiint_T r\,dz\,dr\,d\theta$ in cylindrical coordinates) as your first step and show the innermost z-integration explicitly!

5.2 Worked VIT Question Bank & Tutorial Problems (Topic 5)

Problem 2.5.1 (VIT Question Bank Q14): *Find the volume of the region bounded by the paraboloid $az = x^2 + y^2$ $(a > 0)$, the cylinder $x^2 + y^2 = b^2$, and the plane $z = 0$. Also find the volume when $a = 2$ and $b = 2$.*

Solution:

Here z ranges from the xy-plane $z = 0$ (or if bounded above by $az = x^2 + y^2$ over the disk $x^2 + y^2 \le b^2$) up to $z = \frac{x^2+y^2}{a}$. Setting up the triple integral and converting the xy-projection $(x^2 + y^2 \le b^2)$ to polar/cylindrical coordinates $(x = r\cos\theta, y = r\sin\theta, z = z,\ dV = r\,dz\,dr\,d\theta)$:

$$V = \int_{\theta=0}^{2\pi} \int_{r=0}^{b} \int_{z=0}^{r^2/a} r\,dz\,dr\,d\theta.$$

Step 1 (z-integration):

$$\int_0^{r^2/a} r\,dz = r\left(\frac{r^2}{a}\right) = \frac{r^3}{a}.$$

Step 2 (r and θ integration):

$$V = \int_0^{2\pi} d\theta \int_0^b \frac{r^3}{a}\,dr = 2\pi\left[\frac{r^4}{4a}\right]_0^b = \frac{\pi b^4}{2a}.$$

When $a = 2$ and $b = 2$:

$$V = \frac{\pi(2)^4}{2(2)} = \frac{16\pi}{4} = 4\pi \text{ cubic units.} \quad \square$$

Problem 2.5.2 (VIT Tutorial Sheet Q9): *Find the volume of the cylindrical column standing on the area common to the parabolas $y^2 = x$ and $x^2 = y$, and cut off by the surface $z = 12 + y - x^2$ (and $z = 0$).*

Solution:

The base D in the xy-plane is bounded by $y = x^2$ and $y = \sqrt{x}$, which intersect at $(0, 0)$ and $(1, 1)$. Setting up the triple integral for volume:

$$V = \int_{x=0}^{1} \int_{y=x^2}^{\sqrt{x}} \int_{z=0}^{12+y-x^2} dz\,dy\,dx = \int_0^1 \int_{x^2}^{\sqrt{x}} (12 + y - x^2)\,dy\,dx.$$

Middle y-integration:

$$\begin{aligned}\int_{x^2}^{\sqrt{x}} (12 - x^2 + y)\,dy &= \left[(12 - x^2)y + \frac{y^2}{2}\right]_{y=x^2}^{y=\sqrt{x}} \\ &= 12x^{1/2} + \frac{1}{2}x - 12x^2 - x^{5/2} + \frac{1}{2}x^4.\end{aligned}$$

Outer x-integration:

$$\begin{aligned}V &= \left[8x^{3/2} + \frac{1}{4}x^2 - 4x^3 - \frac{2}{7}x^{7/2} + \frac{1}{10}x^5\right]_0^1 \\ &= 8 + \frac{1}{4} - 4 - \frac{2}{7} + \frac{1}{10} = \frac{569}{140}. \quad \square\end{aligned}$$

Problem 2.5.3 (VIT Tutorial Sheet Q10): *Find the volume bounded by the cylinder $x^2 + y^2 = a^2$ and the planes $y + z = 2a$ and $z = 0$.*

Solution:

Over the circular base $x^2+y^2\le a^2$, z ranges from $z=0$ to $z=2a-y$:

$$V=\iint_{x^2+y^2\le a^2}\int_{z=0}^{2a-y} dz\,dx\,dy=\iint_{x^2+y^2\le a^2}(2a-y)\,dx\,dy.$$

Converting to polar/cylindrical coordinates ($x=r\cos\theta, y=r\sin\theta, dx\,dy=r\,dr\,d\theta$):

$$V=\int_0^{2\pi}\int_0^{a}(2a-r\sin\theta)\,r\,dr\,d\theta=\int_0^{2\pi}\left(a^3-\frac{a^3}{3}\sin\theta\right)d\theta=2\pi a^3. \quad \square$$

6 Topic 6: Change of Variables Between Cartesian and Cylindrical Coordinates

6.1 Theory & Coordinate Relations

Cylindrical coordinates (r,θ,z) combine 2D polar coordinates in the xy-plane with the Cartesian height z:

$$x=r\cos\theta,\qquad y=r\sin\theta,\qquad z=z,\qquad r^2=x^2+y^2.$$

The 3D Jacobian determinant is:

$$J=\frac{\partial(x,y,z)}{\partial(r,\theta,z)}=\begin{vmatrix}\cos\theta & -r\sin\theta & 0\\ \sin\theta & r\cos\theta & 0\\ 0 & 0 & 1\end{vmatrix}=r(\cos^2\theta+\sin^2\theta)=r.$$

Thus the volume element is $\boxed{dV=dx\,dy\,dz=r\,dz\,dr\,d\theta}$, and:

$$\iiint_T f(x,y,z)\,dx\,dy\,dz=\iiint_{T^*} f(r\cos\theta,r\sin\theta,z)\,r\,dz\,dr\,d\theta.$$

6.2 Worked VIT Question Bank & Tutorial Problems (Topic 6)

Problem 2.6.1 (VIT Question Bank Q18 — High-Frequency FAT Problem): *Use a triple integral in cylindrical coordinates to determine the volume of the region that is below the paraboloid $z=8-x^2-y^2$, above the cone $z=-\sqrt{4x^2+4y^2}$, and inside the cylinder $x^2+y^2=4$.*

Solution:

In cylindrical coordinates, $x^2+y^2=r^2$, so:

- Upper surface (paraboloid): $z=8-r^2$.
- Lower surface (inverted cone): $z=-\sqrt{4r^2}=-2r$.
- Cylinder boundary: $r^2\le 4 \implies 0\le r\le 2$ and $0\le\theta\le 2\pi$.

Setting up the triple integral with $dV=r\,dz\,dr\,d\theta$:

$$V=\int_{\theta=0}^{2\pi}\int_{r=0}^{2}\int_{z=-2r}^{8-r^2} r\,dz\,dr\,d\theta.$$

Step 1 (z-integration):

$$\int_{-2r}^{8-r^2} r\,dz=r\Big[(8-r^2)-(-2r)\Big]=8r+2r^2-r^3.$$

Step 2 (r and θ integration):

$$V=\int_0^{2\pi}d\theta\int_0^2\left(8r+2r^2-r^3\right)dr=2\pi\left[4r^2+\frac{2}{3}r^3-\frac{r^4}{4}\right]_0^2=2\pi\left(12+\frac{16}{3}\right)=\frac{104\pi}{3}. \quad \square$$

Problem 2.6.2 (VIT Tutorial Sheet Q7): *Evaluate the triple integral* $I = \int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} dz\,dy\,dx$ *(and compare with the cylindrical version* $\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2}} dz\,dy\,dx$*).*

Solution:

Version A (As printed in VIT Tutorial Sheet Q7): $I = \int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2}} dz\,dy\,dx$. Here the inner z-limit is $\sqrt{9-x^2}$ (independent of y):

$$I = \int_0^3 \int_0^{\sqrt{9-x^2}} \sqrt{9-x^2}\,dy\,dx = \int_0^3 (9-x^2)\,dx = \left[9x - \frac{x^3}{3}\right]_0^3 = 18.$$

Version B (First-octant sphere in cylindrical/spherical coordinates): $I_{\text{sphere}} = \int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} dz\,dy\,dx$ In cylindrical coordinates ($r \in [0,3], \theta \in [0,\pi/2], z \in [0,\sqrt{9-r^2}]$):

$$I_{\text{sphere}} = \int_0^{\pi/2} \int_0^3 \sqrt{9-r^2}\, r\,dr\,d\theta = \frac{\pi}{2}\left[-\frac{1}{3}(9-r^2)^{3/2}\right]_0^3 = \frac{\pi}{2}(9) = \frac{9\pi}{2}. \quad \square$$

7 Topic 7: Change of Variables Between Cartesian and Spherical Coordinates

7.1 Theory & Jacobian Derivation

A point $P(x,y,z)$ in spherical coordinates is represented by (ρ,ϕ,θ) (sometimes denoted (r,θ,ϕ) in Indian textbooks; both notations are identical in meaning):

- $\rho \geq 0$: Radial distance from the origin, $\rho = \sqrt{x^2+y^2+z^2}$.
- $\phi \in [0,\pi]$: Angle from the positive z-axis (colatitude), so $z = \rho\cos\phi$ and $r_{xy} = \sqrt{x^2+y^2} = \rho\sin\phi$.
- $\theta \in [0,2\pi]$: Azimuthal angle in the xy-plane from the positive x-axis.

Thus:

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi, \qquad x^2+y^2+z^2 = \rho^2.$$

Computing the Jacobian $\frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)}$ yields $|J| = \rho^2\sin\phi$, giving the spherical volume element:

$$\boxed{dV = dx\,dy\,dz = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta.}$$

7.2 Worked VIT FAT Past Paper Problems (Topic 7)

Problem 2.7.1 (BMAT101L FAT November 2024 Slot A1): *Evaluate* $\iiint_V \frac{dx\,dy\,dz}{\sqrt{1-x^2-y^2-z^2}}$ *over the positive octant of the sphere* $x^2+y^2+z^2 = 1$ *(and over the entire sphere) by transforming into spherical polar coordinates.*

Solution:

In spherical coordinates, $x^2+y^2+z^2 = \rho^2$ and $dx\,dy\,dz = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$. For the **positive octant** ($x \geq 0, y \geq 0, z \geq 0$) of the unit sphere $\rho \leq 1$:

$$0 \leq \rho \leq 1, \qquad 0 \leq \phi \leq \frac{\pi}{2}, \qquad 0 \leq \theta \leq \frac{\pi}{2}.$$

Thus:

$$I_{\text{octant}} = \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{\pi/2} \int_{\rho=0}^{1} \frac{\rho^2\sin\phi}{\sqrt{1-\rho^2}}\,d\rho\,d\phi\,d\theta = \left(\int_0^{\pi/2} d\theta\right)\left(\int_0^{\pi/2} \sin\phi\,d\phi\right)\left(\int_0^1 \frac{\rho^2}{\sqrt{1-\rho^2}}\,d\rho\right).$$

Let $\rho = \sin t \implies d\rho = \cos t\, dt$, with $t \in [0, \pi/2]$:

$$\int_0^1 \frac{\rho^2}{\sqrt{1-\rho^2}}\, d\rho = \int_0^{\pi/2} \frac{\sin^2 t}{\cos t} \cos t\, dt = \int_0^{\pi/2} \sin^2 t\, dt = \frac{\pi}{4}.$$

Since $\int_0^{\pi/2} d\theta = \frac{\pi}{2}$ and $\int_0^{\pi/2} \sin\phi\, d\phi = 1$:

$$I_{\text{octant}} = \left(\frac{\pi}{2}\right)(1)\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8}.$$

For the **entire sphere** ($\phi \in [0, \pi], \theta \in [0, 2\pi]$), multiply by 8: $I_{\text{full}} = 8 \times \frac{\pi^2}{8} = \pi^2$. □

Problem 2.7.2 (VIT FAT Repeated Classic — Ice-Cream Cone Volume): *Using spherical coordinates, find the volume of the solid bounded above by the sphere $x^2 + y^2 + z^2 = a^2$ and below by the cone $z = \sqrt{x^2 + y^2}$.*

Solution:

1. **Sphere boundary**: $x^2 + y^2 + z^2 = a^2 \implies \rho = a$, so $0 \le \rho \le a$. 2. **Cone boundary**: $z = \sqrt{x^2 + y^2} \implies \rho\cos\phi = \rho\sin\phi \implies \tan\phi = 1 \implies \phi = \frac{\pi}{4}$. Since the solid lies above the cone around the positive z-axis, $0 \le \phi \le \frac{\pi}{4}$. 3. **Azimuthal angle**: Full revolution around the z-axis gives $0 \le \theta \le 2\pi$. Therefore:

$$V = \int_{\theta=0}^{2\pi}\int_{\phi=0}^{\pi/4}\int_{\rho=0}^{a} \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta = \left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi/4} \sin\phi\, d\phi\right)\left(\int_0^a \rho^2\, d\rho\right)$$
$$= (2\pi)\left(1 - \frac{1}{\sqrt{2}}\right)\left(\frac{a^3}{3}\right) = \frac{\pi a^3}{3}\left(2 - \sqrt{2}\right). \quad \square$$

8 Topic 8: Change of Variables for Object Recognition

8.1 Theory: General Jacobian Transformations & Shape Normalization

In general 2D coordinate transformations $(x, y) \leftrightarrow (u, v)$:

$$\iint_R f(x,y)\, dx\, dy = \iint_{R^*} f(x(u,v), y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\, dv, \qquad \text{where} \quad \frac{\partial(x,y)}{\partial(u,v)} = \left(\frac{\partial(u,v)}{\partial(x,y)}\right)^{-1}.$$

Why This Matters in Object Recognition (Computer Vision): When a camera captures an object at an angle or different distance, the image undergoes an **affine transformation** (rotation, scaling, shearing, and translation):

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}, \qquad \frac{\partial(u,v)}{\partial(x,y)} = a_{11}a_{22} - a_{12}a_{21} = \det(A).$$

By applying a change of variables $(x, y) \to (u, v)$, an arbitrarily rotated or skewed bounding box (such as a tilted diamond, skewed parallelogram, or stretched ellipse) is mapped to a **canonical unit square** $[-1, 1]^2$ or **unit disk** $u^2 + v^2 \le 1$, enabling scale- and rotation-invariant template matching!

8.2 Worked VIT Question Bank Problems (Topic 8)

Problem 2.8.1 (VIT Question Bank Q12 — Rotated Square Transformation): *Evaluate $\iint_R (1 - x^2 - y^2)\, dx\, dy$, where R is the region bounded by the square with vertices $(\pm 1, 0)$ and $(0, \pm 1)$, using the change of coordinates $u = x - y$, $v = x + y$.*

Solution:

Step 1 (Find new limits in (u,v)): The four edges of the diamond-shaped square with vertices $(1,0),(0,1),(-1,0),(0,-1)$ are:

$$x-y=1,\quad x-y=-1,\quad x+y=1,\quad x+y=-1 \implies -1\le u\le 1 \quad\text{and}\quad -1\le v\le 1.$$

Step 2 (Express integrand and Jacobian in (u,v)): Since $x=\frac{u+v}{2}$ and $y=\frac{v-u}{2}$, we have $x^2+y^2=\frac{(u+v)^2+(v-u)^2}{4}=\frac{2(u^2+v^2)}{4}=\frac{u^2+v^2}{2}$. The Jacobian is:

$$\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix}1 & -1\\ 1 & 1\end{vmatrix}=2 \implies \left|\frac{\partial(x,y)}{\partial(u,v)}\right|=\frac{1}{2}.$$

Step 3 (Integrate over the canonical square $[-1,1]\times[-1,1]$):

$$I=\frac{1}{2}\int_{-1}^{1}\int_{-1}^{1}\left(1-\frac{u^2+v^2}{2}\right)du\,dv=\frac{1}{2}\int_{-1}^{1}\left(\frac{5}{3}-v^2\right)dv=\frac{1}{2}\left[\frac{5}{3}v-\frac{v^3}{3}\right]_{-1}^{1}=\frac{4}{3}. \quad \square$$

Problem 2.8.2 (VIT Question Bank Q19 — Parallelogram Change of Variables): *Evaluate $\iint_D xe^{x^2-y^2}\,dx\,dy$ where D is the closed region bounded by the lines $y=x$, $y=x-1$, $y=0$, and $y=1$.*

Solution:

Method 1 (Direct Horizontal Strips $dx\,dy$): Notice that for fixed $y\in[0,1]$, x ranges from $x=y$ to $x=y+1$, and $xe^{x^2-y^2}\,dx$ integrates directly via $w=x^2-y^2 \implies dw=2x\,dx$!

$$\begin{aligned}I&=\int_{y=0}^{1}\int_{x=y}^{y+1}xe^{x^2-y^2}\,dx\,dy=\int_0^1\left[\frac{1}{2}e^{x^2-y^2}\right]_{x=y}^{x=y+1}dy\\&=\frac{1}{2}\int_0^1\left(e^{2y+1}-1\right)dy=\frac{1}{4}(e^3-e)-\frac{1}{2}.\end{aligned}$$

Method 2 (Using Jacobian Change of Variables $u=x-y, v=y$): Here $u\in[0,1]$, $v\in[0,1]$, $x=u+v$, $x^2-y^2=(x-y)(x+y)=u(u+2v)$, and $|J|=1$, giving the exact same result $\frac{1}{4}(e^3-e)-\frac{1}{2}$. $\square$

9 Topic 9: Multiple Integrals in Image Analysis for Density and Moments

9.1 Theory: Image Intensity, Mass, Centroid, and Central/Inertia Moments

In continuous 2D image processing, a grayscale image is modeled as a non-negative density/intensity function $I(x,y)$ (or $\rho(x,y)$) over a 2D spatial domain R:

1. **Zeroth Moment (Total Mass / Total Image Intensity M_{00}):**
$$M=M_{00}=\iint_R \rho(x,y)\,dx\,dy.$$
2. **First Raw Moments (M_{10},M_{01}) and Centroid / Center of Gravity $(\bar{x},\bar{y})$:**
$$M_{10}=\iint_R x\,\rho(x,y)\,dx\,dy,\qquad M_{01}=\iint_R y\,\rho(x,y)\,dx\,dy,\qquad \bar{x}=\frac{M_{10}}{M_{00}},\qquad \bar{y}=\frac{M_{01}}{M_{00}}.$$
3. **General (p,q)-Order Raw Moment and Central Moment (Translation Invariant):**
$$M_{pq}=\iint_R x^p y^q\rho(x,y)\,dx\,dy,\qquad \mu_{pq}=\iint_R (x-\bar{x})^p(y-\bar{y})^q\rho(x,y)\,dx\,dy.$$
4. **Second Moments / Moments of Inertia (I_x,I_y,I_0) and Principal Orientation Angle θ:**
$$I_x=\iint_R y^2\rho(x,y)\,dx\,dy,\qquad I_y=\iint_R x^2\rho(x,y)\,dx\,dy,\qquad I_0=I_x+I_y=\iint_R (x^2+y^2)\rho(x,y)\,dx\,dy.$$
The orientation angle θ of the object's major axis in image analysis is given by $\tan(2\theta)=\frac{2\mu_{11}}{\mu_{20}-\mu_{02}}$.

9.2 Worked Applied Problems (Topic 9)

Problem 2.9.1 (Applied FAT Question — Image Density, Centroid & Polar Moment): *A circular object in a segmented grayscale image occupies the first-quadrant quadrant disk $R : x^2 + y^2 \le a^2,\ x \ge 0,\ y \ge 0$, with pixel intensity (density) proportional to the distance from the origin: $\rho(x,y) = k\sqrt{x^2+y^2}$. Find: (i) the total intensity (mass) M_{00}, (ii) the centroid $(\bar{x},\bar{y})$, and (iii) the moments of inertia I_x, I_y, and I_0.*

Solution:

Convert to polar coordinates: $r \in [0,a]$, $\theta \in [0,\pi/2]$, $\rho(r,\theta) = kr$, and $dx\,dy = r\,dr\,d\theta$.

1. **Total Mass / Intensity** M_{00}:

$$M_{00} = \int_0^{\pi/2}\int_0^a (kr)\,r\,dr\,d\theta = k\left(\frac{\pi}{2}\right)\left[\frac{r^3}{3}\right]_0^a = \frac{\pi k a^3}{6}.$$

2. **First Moments and Centroid** $(\bar{x},\bar{y})$:

$$M_{10} = \int_0^{\pi/2}\int_0^a (r\cos\theta)(kr)\,r\,dr\,d\theta = k\Big[\sin\theta\Big]_0^{\pi/2}\left[\frac{r^4}{4}\right]_0^a = \frac{ka^4}{4}.$$

   By symmetry, $M_{01} = \frac{ka^4}{4}$. Therefore:

$$\bar{x} = \bar{y} = \frac{M_{10}}{M_{00}} = \frac{ka^4/4}{\pi k a^3/6} = \frac{3a}{2\pi}.$$

3. **Moments of Inertia** I_x, I_y, I_0:

$$I_0 = \iint_R (x^2+y^2)\rho(x,y)\,dx\,dy = \int_0^{\pi/2}\int_0^a r^2(kr)\,r\,dr\,d\theta = k\left(\frac{\pi}{2}\right)\left[\frac{r^5}{5}\right]_0^a = \frac{\pi k a^5}{10}.$$

   By symmetry between x and y in the first quadrant, $I_x = I_y = \frac{1}{2}I_0 = \frac{\pi k a^5}{20}$. □

10 Additional VIT Question Bank Problems & Self-Test Practice

Problem 2.10.1 (VIT Question Bank Q6): *Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} \log(x^2+y^2)\,dx\,dy$ $(a>0)$ by converting to polar coordinates.*

Solution:

The region $0 \le x \le \sqrt{a^2-y^2}$, $0 \le y \le a$ is the first-quadrant sector of the disk $x^2+y^2 \le a^2$, so $0 \le r \le a$ and $0 \le \theta \le \frac{\pi}{2}$. In polar coordinates, $\log(x^2+y^2) = \log(r^2) = 2\log r$, and $dx\,dy = r\,dr\,d\theta$:

$$I = \int_0^{\pi/2}\int_0^a (2\log r)\,r\,dr\,d\theta = \frac{\pi}{2}\int_0^a 2r\log r\,dr.$$

Using integration by parts ($u = \log r, dv = 2r\,dr \implies du = \frac{1}{r}dr, v = r^2$):

$$\int_0^a 2r\log r\,dr = \left[r^2\log r - \int r^2\cdot\frac{1}{r}\,dr\right]_0^a = \left[r^2\log r - \frac{r^2}{2}\right]_0^a = a^2\log a - \frac{a^2}{2}.$$

Multiplying by $\frac{\pi}{2}$ gives:

$$I = \frac{\pi a^2}{2}\left(\log a - \frac{1}{2}\right). \quad \square$$

Problem 2.10.2 (VIT Question Bank Q8): *Evaluate* $\int_0^a \int_{\sqrt{ax-x^2}}^{\sqrt{a^2-x^2}} \frac{1}{\sqrt{a^2-x^2-y^2}}\, dy\, dx$ *by changing into polar coordinates.*

Solution:

Here $y = \sqrt{ax - x^2} \iff x^2 + y^2 = ax \iff r = a\cos\theta$, and $y = \sqrt{a^2 - x^2} \iff x^2 + y^2 = a^2 \iff r = a$, in the first quadrant ($\theta \in [0, \pi/2]$). Thus:

$$I = \int_0^{\pi/2} \int_{a\cos\theta}^{a} \frac{r\, dr}{\sqrt{a^2 - r^2}}\, d\theta = \int_0^{\pi/2} \Big[-\sqrt{a^2 - r^2}\Big]_{r=a\cos\theta}^{r=a}\, d\theta = \int_0^{\pi/2} a\sin\theta\, d\theta = a\Big[-\cos\theta\Big]_0^{\pi/2} = a.$$

(Note: If the integrand is $\sqrt{x^2 + y^2} = r$, then $\int_0^{\pi/2} \int_{a\cos\theta}^{a} r^2\, dr\, d\theta = \frac{a^3}{3} \int_0^{\pi/2} (1 - \cos^3\theta)\, d\theta = \frac{a^3}{3}\left(\frac{\pi}{2} - \frac{2}{3}\right)$.)

□

Quick Self-Test Checklist Before Your Exam:

1. Did you multiply by the Jacobian (r for polar/cylindrical, $\rho^2 \sin\phi$ for spherical, $\left|\frac{\partial(x,y)}{\partial(u,v)}\right|$ for general (u, v))?
2. In spherical coordinates, did you check that ϕ stops at $\pi/2$ for $z \geq 0$ and π for a full sphere?
3. When asked to **change the order of integration**, did you draw the region and verify that the new outer limits are constants?